

Predicting Clothing Fit:

A Machine Learning Approach to Size Recommendation

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Instructor: Anil Chaturvedi

Authors: Harleen Kaur Buttar | Skylar Liu | Dora Jiayue Li

Github Repository: [Our code here!](#)

Executive Summary

Online apparel retail loses an estimated 30–40% of revenue to returns, with poor fit as the leading cause. Standard sizing systems cannot account for the continuous variation in body types, the contextual nature of garment fit, or the subjective dimension of how customers perceive fit. This project builds an end-to-end machine learning pipeline that addresses all three failure modes simultaneously, and in doing so, creates a scalable infrastructure for personalized size recommendation.

Key Finding

KNN trained on the rich RentTheRunway feature set achieves 90.7% accuracy, 0.869 macro-F1, and a 26% improvement over the same model trained on basic measurements alone. The richness of the feature engineering, not just the choice of algorithm, drives predictive performance.

Strategic Implications by Stakeholder

For Retail Operations & Merchandising

The pipeline identifies a meaningful minority of customers (flagged by DBSCAN as outliers and by GMM as low-confidence assignments) who are systematically underserved by standard sizing. These customers represent the highest return risk and the highest opportunity for targeted intervention. Prioritizing personalized recommendations for this segment directly reduces return rates and associated logistics costs.

For Product & UX Teams

LDA topic modeling on 176,562 reviews identified 11 coherent fit-language themes. Reviews labeled Small or Large show disproportionately higher proportions of the sizing discussion topic, while Fit reviews skew toward positive social experience language. This signals that customer language at point-of-review is a leading indicator of fit quality and that it can be incorporated into real-time feedback loops and product description optimization.

For Data Science & Engineering

Unsupervised feature discovery feeding supervised prediction feeding a collaborative recommender in a modular pipeline architecture designed for generalization. Cross-retailer evaluation (train on ModCloth, test on RentTheRunway) demonstrates that body-type and fit patterns transfer across brands when operating at the category level, achieving Hit@1 of 10.5% and Hit@3 of 32.2%. This establishes a foundation for multi-retailer deployment.

For Executive Leadership

The project demonstrates that fit prediction is a solvable problem with available data. The pipeline is not a single model but a layered system consisting of clustering, classification, and recommendation, with each component adding measurable lift. The key investment required is in data richness: body measurements, review text, and garment context together generate the feature space that makes accurate prediction possible. Retailers that invest in collecting these signals will have a compounding advantage over those relying on static size charts.

Datasets

Three datasets support the project pipeline, each serving a distinct purpose across the three phases. Together they provide complementary coverage of customer body measurements, garment context, and fit outcomes spanning two real-world apparel retailers and one clean benchmark dataset.

Dataset	Transactions	Primary Role	Key Features
RentTheRunway (PRIMARY)	192,544	All three phases	Fit feedback, body measurements, review text, occasion context
ModCloth (SECONDARY)	82,790	Phase 3 cross-retailer generalisation	Fit feedback, hips, bra/cup size, category, length feedback
Clothes-Size-Prediction (SUPPLEMENT)	118,871	Phase 2 benchmark baseline	Weight, age, height → size (XXS–XXXL); fully complete, no missing values

Dataset 1 — RentTheRunway (PRIMARY) | 192,544 transactions

RentTheRunway is the richest dataset and serves as the foundation for all three phases. It contains structured body measurements, fit feedback, rental occasion context, product ratings, and full review text, providing every signal needed for unsupervised exploration, supervised classification, and size recommendation.

Features

- **Fit outcome label:** Fit feedback:
 - Small / Fit / Large (three-class target throughout the pipeline)
- **Continuous predictors:** Body measurements:
 - Height, weight, bust size, age, body type
- **Unstructured and structured review signals:** Review data:
 - Review text (used for LDA topic modeling), summary, star rating, submission date
- **Categorical rental attributes:** Garment context:
 - Clothing category (dress, top, etc.), rented-for occasion (wedding, work, everyday, etc.)

Why RentTheRunway was chosen as primary

ModCloth had critical feature sparsity. Waist and bust measurements were over 80% null, body type was absent entirely, and shoe size/width fields (66–78% null) added noise without predictive value.

RentTheRunway had far fewer missing values across all key body measurement fields and uniquely offered full review text, occasion context, and body type — making it the only dataset capable of supporting all three phases.

Dataset 2 — ModCloth (SECONDARY) | 82,790 transactions

ModCloth provides fit feedback from a second retailer with a partially overlapping but distinct measurement schema. Despite high sparsity on several columns, it was retained for its value in cross-retailer generalization testing in Phase 3 to assess whether fit patterns learned from RentTheRunway transfer to a different brand and customer base.

Features retained after cleaning

- Fit feedback: Small / Fit / Large
- Body measurements: height (inches), hips, bra size, cup size — high sparsity on hips and cup size
- Review data: review text, summary, quality rating
- Garment context: clothing category, length feedback

Features dropped due to sparsity

- Waist and bust measurements (> 80% null): dropped entirely
- Shoe size and shoe width (66–78% null): irrelevant to garment fit and dropped
- Body type: not collected by ModCloth; absent from all records

Dataset 3 — Clothes-Size-Prediction (SUPPLEMENT) | 118,871 rows

The Clothes-Size-Prediction dataset is a clean, fully complete benchmark containing only three features (weight, age, and height) mapped to a clothing size label (XXS through XXXL). It contains no missing values and no review or occasion data. Its sole purpose in the pipeline is to serve as a controlled baseline for Phase 2: by training all seven classifiers on this minimal feature set and comparing against the same models trained on the rich RentTheRunway features, the incremental predictive value of the Phase I feature engineering can be precisely quantified.

Features

- Body measurements: weight (kg), age, height (cm), with conversions to imperial units added during cleaning

- Target: clothing size (XXS / XS / S / M / L / XL / XXL / XXXL), plus a mapped fit equivalence (Small / Fit / Large) for cross-dataset comparison
- Engineered features: BMI, height in inches, weight in pounds (added for consistency with the RentTheRunway pipeline)
- Zero missing values across all rows and columns

Data source

- Datasets 1 & 2: [Misra et al., Kaggle]

(<https://www.kaggle.com/datasets/rmisra/clothing-fit-dataset-for-size-recommendation>)

- Dataset 3: [Dubey, Kaggle]

(<https://www.kaggle.com/datasets/blueist55/clothesizeprediction>)

Data Cleaning & Preparation

Data cleaning for this project was shaped by two competing priorities: preserving statistical validity for the unsupervised phase, and maximising sample size for the supervised phase. These goals required different missing-data strategies at each stage of the pipeline.

Why RentTheRunway Was Selected as the Primary Dataset

The cleaning process began with a systematic sparsity comparison across the three datasets. Despite its 82,790 transactions, ModCloth was disqualified as a primary source by the extent of missing data on the features most critical to fit prediction.

Feature	RTR Missing %	ModCloth Missing %	Decision
Waist / Bust	< 5%	> 80%	Dropped from ModCloth; retained in RTR
Body Type	Low	N/A	RTR primary for clustering
Hips	N/A	32.3%	High sparsity; excluded from Phase 1 clustering
Cup Size	Low	32.8%	Used in RTR; dropped in ModCloth pipeline
Shoe Size / Width	N/A	66–78%	Dropped, irrelevant to garment fit
Review Text	Low	8.1%	RTR used for LDA; ModCloth excluded from topic modeling

With waist and bust missing in over 80% of ModCloth records, body type absent entirely, and shoe-related fields (66–78% null) adding noise without signal, ModCloth could not support unsupervised body-type clustering or supervised fit prediction. RentTheRunway had materially lower missingness across all key fields and uniquely provided review text, occasion context, and self-reported body type, making it the only dataset capable of supporting all three pipeline phases.

Missing Data Strategy — Phase 1 (Unsupervised)

During the unsupervised feature discovery phase (DBSCAN, Gaussian Mixture Models, K-Means, and K-Modes) observations with missing measurement values were excluded prior to model fitting. This was a deliberate methodological choice, not a convenience.

Why complete-case analysis was used for clustering

Distance-based and likelihood-based methods (K-Means, GMM, DBSCAN) require well-defined positions in the feature space. Imputing missing values at this stage would artificially distort the geometry of the data, forcing observations into clusters based on imputed values that do not reflect actual body measurements. Restricting clustering to complete cases ensured that latent body-type segments were learned from valid, comparable feature vectors, producing more meaningful and stable cluster assignments.

Missing Data Strategy — Phase 2 (Supervised)

For downstream supervised modeling, a different strategy was appropriate. Cluster assignments and LDA topic features from Phase 1 were treated as derived inputs rather than core measurements. Because the unsupervised models were trained on different subsets of the full dataset, merging their outputs back into the complete transaction table naturally reintroduced missing values for observations that had been excluded from clustering.

Rather than discarding these observations, missing values were imputed within the modeling pipeline using strategies matched to each feature type:

- Numeric features (body measurements, PCA components, topic probabilities): median imputation
- Categorical features (cluster labels, garment category, occasion): most-frequent value imputation
- Binary indicators (DBSCAN outlier flag): constant imputation (0 = not an outlier)

This approach preserved sample size, avoided selection bias toward customers with complete records, and produced models capable of making predictions under realistic deployment conditions where some customer information may be unavailable.

RentTheRunway — Cleaned Dataset Schema

After cleaning, the RentTheRunway dataset contains 23 columns spanning raw measurements, engineered features, garment context, review signals, and temporal attributes. All columns used in downstream modeling are documented below.

Column	Type	Description
user_id	Identifier	Unique customer identifier
item_id	Identifier	Unique product identifier
fit	Target	Raw fit label (small / fit / large)
fit_label	Target	Numeric encoding: 0 = Small, 1 = Fit, 2 = Large
height_inches	Body Measurement	Customer height converted to inches
weight_lbs	Body Measurement	Customer weight in pounds
bmi	Engineered	Calculated from height and weight
bust_band	Body Measurement	Numeric bust band size
cup_size_letter	Body Measurement	Original cup size letter (A–K)
cup_size_num	Engineered	Numeric encoding of cup size letter
size	Garment	Numeric garment size ordered
age	Body Measurement	Customer age
category	Garment	Clothing category (e.g., dress, top)
body_type	Body Profile	Self-reported body type (e.g., hourglass, petite)
body_type_num	Engineered	Numeric encoding of body type
rented_for	Context	Occasion (e.g., wedding, work, everyday)
rented_for_num	Engineered	Numeric encoding of occasion
rating	Review	Numeric product rating (1–10)
review_summary	Review	Short review headline text
review_text	Review	Full review body (used for LDA)
review_date	Temporal	Date of review submission
review_year	Temporal	Extracted year
review_month	Temporal	Extracted month

Engineered columns added during cleaning

- **bmi**: calculated from height and weight for all records; used as a composite body measurement in clustering and classification

- `cup_size_num`: ordinal numeric encoding of cup size letters (A = 1 through K = 11) to enable use in distance-based models
- `body_type_num`, `rented_for_num`, `fit_label`: numeric encodings of categorical fields for compatibility with all seven classifiers
- `review_year`, `review_month`: extracted from `review_date` to enable temporal analysis

Clothes-Size-Prediction — Cleaned Dataset Schema

The Clothes-Size-Prediction dataset required minimal cleaning — it contained no missing values. The primary cleaning work involved unit conversion and target variable engineering to enable direct comparison with the RentTheRunway pipeline.

Column	Type	Description
<code>weight_kg</code>	Body Measurement	Weight in kilograms (original)
<code>age</code>	Body Measurement	Customer age
<code>height_cm</code>	Body Measurement	Height in centimetres (original)
<code>size</code>	Target	Clothing size label (XXS / XS / S / M / L / XL / XXL / XXXL)
<code>size_label</code>	Target	Numeric encoding of size (1–7)
<code>fit_equiv</code>	Engineered	Mapped fit equivalence (Small / Fit / Large)
<code>fit_equiv_label</code>	Engineered	Numeric encoding: 0 = Small, 1 = Fit, 2 = Large
<code>height_inches</code>	Engineered	Height converted to inches for cross-dataset compatibility
<code>weight_lbs</code>	Engineered	Weight converted to pounds for cross-dataset compatibility
<code>bmi</code>	Engineered	Calculated from height and weight

Engineered columns added during cleaning

- `height_inches`, `weight_lbs`: converted from metric for consistency with the RentTheRunway feature space
- `bmi`: calculated from height and weight, consistent with RentTheRunway
- `fit_equiv` / `fit_equiv_label`: sizes S/XS/XXS mapped to Small, M mapped to Fit, L/XL/XXL/XXXL mapped to Large, enabling direct comparison of classifier performance across datasets

Phase I — Unsupervised Exploration

Objective

Predicting clothing fit requires understanding two distinct but equally important dimensions of why fit fails: who the customer is and what they are renting. Phase I transforms raw transaction data into a rich feature space by discovering latent structure across body profiles, garment context, and customer language, making it possible to predict whether a garment will fit before a customer ever tries it on.

Why Multiple Clustering Methods Were Necessary

Fit failure is multidimensional. No single algorithm captures all the relevant structure simultaneously, because the data contains continuous body measurements, categorical garment attributes, probabilistic overlap between body types, and natural outliers, each requiring a different modeling approach.

Method	Input Type	Business Role
K-Means	Continuous body measurements	Clean, interpretable body-type segments for downstream feature engineering
GMM	Continuous body measurements	Probabilistic cluster memberships reflecting that body types exist on a spectrum
DBSCAN	Continuous body measurements	Outlier detection → flags customers standard sizing fails most
K-Modes	Categorical (garment type, occasion, fit feedback)	Isolates behavioral and contextual rental patterns invisible to numeric methods
LDA	Review text (176,562 reviews)	Extracts latent fit-language themes from unstructured customer language

Key Findings from Clustering

Body measurements are continuous, not discrete

Across all clustering methods, no sharp boundaries emerged between body types. Most customers fall within a broad continuum, which is precisely why fixed sizing systems fail. Soft assignments (GMM confidence scores, DBSCAN outlier flags) are more informative than hard labels alone because they preserve the uncertainty inherent in body measurement data.

Categorical context matters as much as physical measurements

K-Modes revealed that fit complaints cluster around specific combinations of garment category, body type, and rental occasion (patterns entirely invisible to numeric clustering). A customer's fit outcome is shaped by what they are renting and why, not just their measurements. A floor-length gown and a bodycon dress fit differently on the same body.

A meaningful minority are consistently mis-served

Every clustering method surfaced a subset of users whose body profiles sit outside the mainstream. DBSCAN noise points, GMM low-confidence assignments, and K-Means' smallest clusters all point to the same population: customers most likely to report poor fit, and the highest-value targets for personalized sizing recommendations.

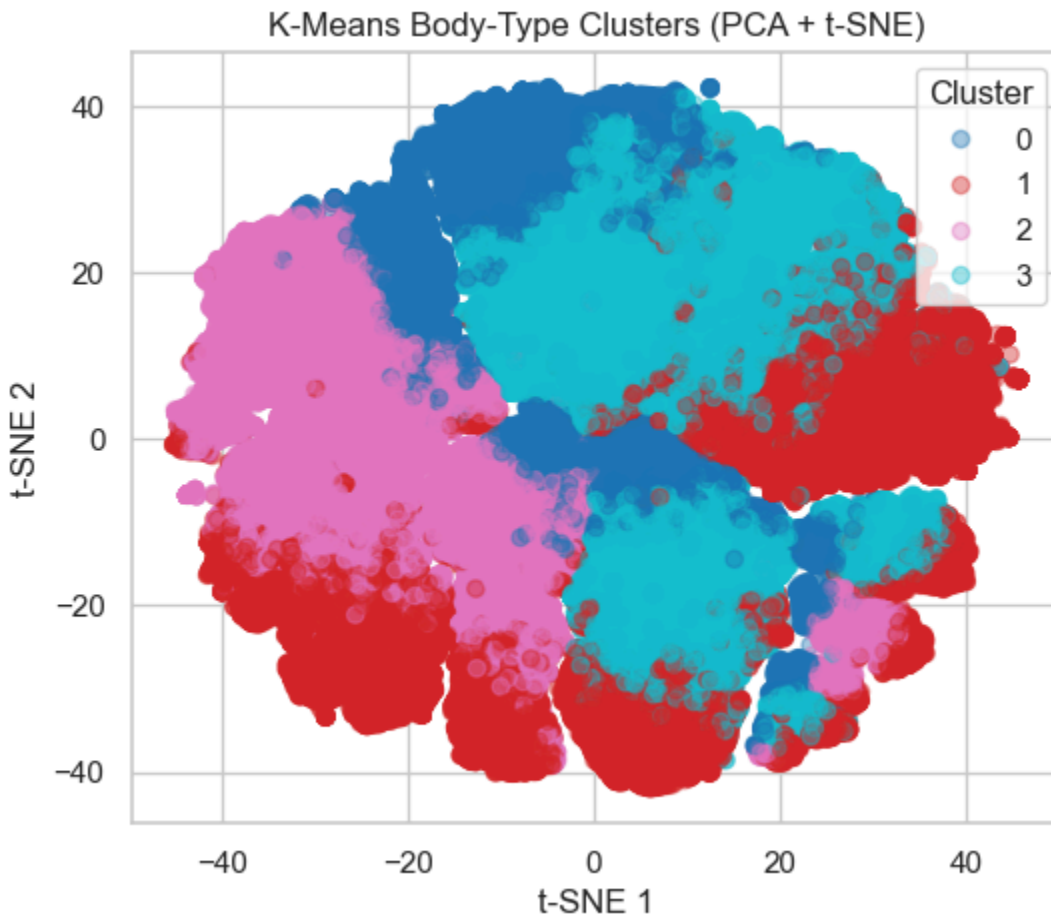


Figure 1: K-Means Body Type Clusters (PCA + t-SNE)

The visualization shows K-Means body-type clusters projected with PCA and t-SNE. The heavy overlap between clusters indicates that body measurements form a **continuous spectrum rather than distinct groups**, reinforcing why fixed sizing systems often fail to capture real body diversity.

LDA Topic Modeling

Customer review text carries fit information that body measurements and garment categories cannot encode. A customer might cluster neatly into a petite segment yet report 'tight across shoulders' or 'runs small' (signals that only appear in language). LDA was applied to extract this latent semantic layer and make it available as structured features for Phase II classification.

LDA Results

11 coherent topics identified from 176,562 reviews (k selected by highest coherence score, c_v). Topics spanned garment fit, fabric and construction, styling occasions, and user experience. Reviews labeled Small or Large showed higher proportions of the sizing discussion topic; Fit reviews contained more positive social experience language. These topic probability distributions were incorporated as features in Phase II.

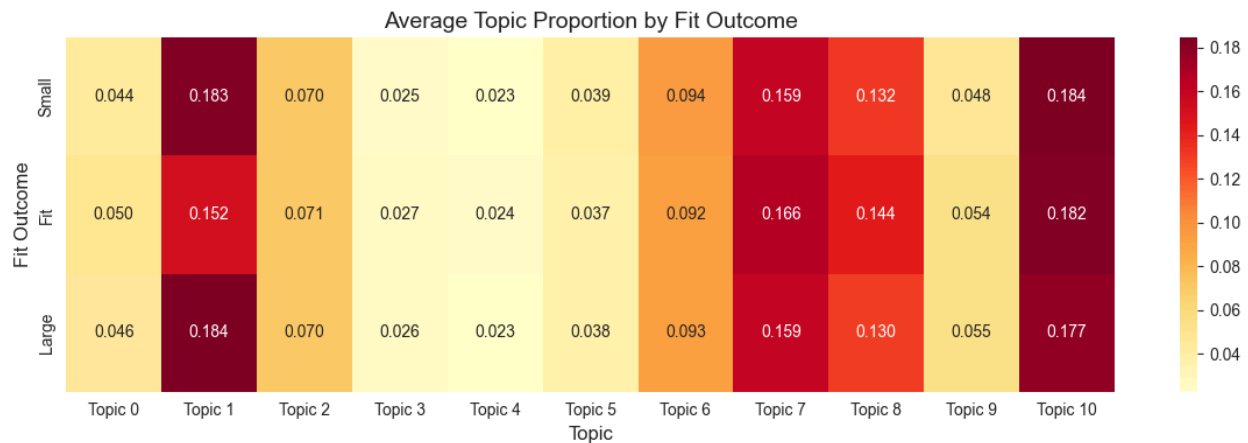


Figure 2: Average Topic Proportion by Fit Outcome

The heatmap shows average topic proportions across fit outcomes. Topics related to **sizing discussion appear more frequently in Small and Large reviews**, while **Fit reviews contain more general positive experience language**, confirming that review text provides useful signals for predicting fit outcomes.

Phase I Outputs — Saved for Phase II

All unsupervised outputs were saved as structured features for downstream supervised modeling:

- Dataframes with cluster assignments (K-Means, GMM, DBSCAN labels)
- PCA components from the body measurement space
- t-SNE visualizations for exploratory validation
- Cluster profiles summarizing segment characteristics
- LDA topic assignments and per-review topic probability distributions

These outputs serve a dual purpose: they enrich the feature space for Phase II classifiers, and they constitute a standalone segmentation of customer body types and rental behavior with direct business value.

Phase II — Supervised Fit Prediction

Objective

Predict fit outcome (Small / Fit / Large) as a multiclass classification problem. The classifier operationalizes the Phase I feature engineering, turning latent body segments, garment context clusters, and review topics into a reliable prediction of whether a garment will fit a specific customer.

Why Seven Models Were Evaluated in Parallel

Fit prediction is driven by a mix of continuous measurements, categorical garment features, and probabilistic text signals. No single algorithm handles all of these input types equally well. Running multiple model families ensures the pipeline captures every dimension of the problem, and cross-validating all seven allows the best-performing model to be selected with confidence.

Model	Strength	Role in This Pipeline
CART	Interpretable decision boundaries	Baseline: mirrors how a stylist reasons about fit; easy to explain to stakeholders
Random Forest	Variance reduction via ensembling	Handles noisy body measurement data; smooths out individual tree overfitting
Bagging	Ensemble stability	Reduces variance on diverse body types; complements Random Forest
Kernel SVM	Non-linear boundaries	Captures complex interactions between body measurements and topic distributions
Linear SVM	Linear separability	Tests whether fit classes are linearly separable in the engineered feature space
KNN	Reasoning by analogy	Directly mirrors personalized recommendation logic: similar customers, similar fit outcomes
Naive Bayes	Probabilistic baseline	Handles LDA topic probability distributions naturally; fast probabilistic baseline

Feature Engineering

The rich feature set combined three categories of signals:

- Raw body measurements: height, weight, bust, BMI, age
- Cluster-derived features: GMM membership probabilities, DBSCAN outlier flags, K-Means segment labels
- LDA topic distributions: 11 topic probability scores per review

A unified ColumnTransformer pipeline applied median imputation and standardization for numeric features, one-hot encoding for categorical cluster labels, and pass-through for binary indicators, which ensures consistent preparation across all seven models and prevents data leakage within cross-validation folds.

Model Performance Results

Model	Accuracy	Macro F1	Notable Characteristic
KNN (Rich Features)	90.7%	0.869	Best overall — exploits similarity structure in rich feature space
Kernel SVM	82.0%	0.780	Strong non-linear boundary; improved with rich features
Random Forest	77.0%	~0.74	Consistent; moderate improvement from rich features
Bagging	75.0%	~0.72	Stable ensemble baseline
CART	73.0%	~0.70	Interpretable; underperforms with expanded feature space
Naive Bayes	~65%	~0.61	Violated independence assumptions hurt performance
Linear SVM	~60%	~0.50	Defaulted to majority Fit class; feature space too complex

The Value of Rich Features: Benchmark Comparison

To quantify the value of the Phase I feature engineering, all seven models were also trained on a simple baseline dataset containing only weight, age, and height. The performance delta reveals which model families benefit most from richer representation.

Key Finding: Feature Richness Drives KNN Performance

KNN macro-F1 improves from 0.677 (simple features) to 0.869 (rich features) — a 28% gain. Kernel SVM improves from 0.686 to 0.780. Ensemble models see smaller but consistent gains. Linear SVM, CART, and Naive Bayes perform worse with richer features, as the expanded feature space introduces correlations and non-linear interactions these models cannot handle.

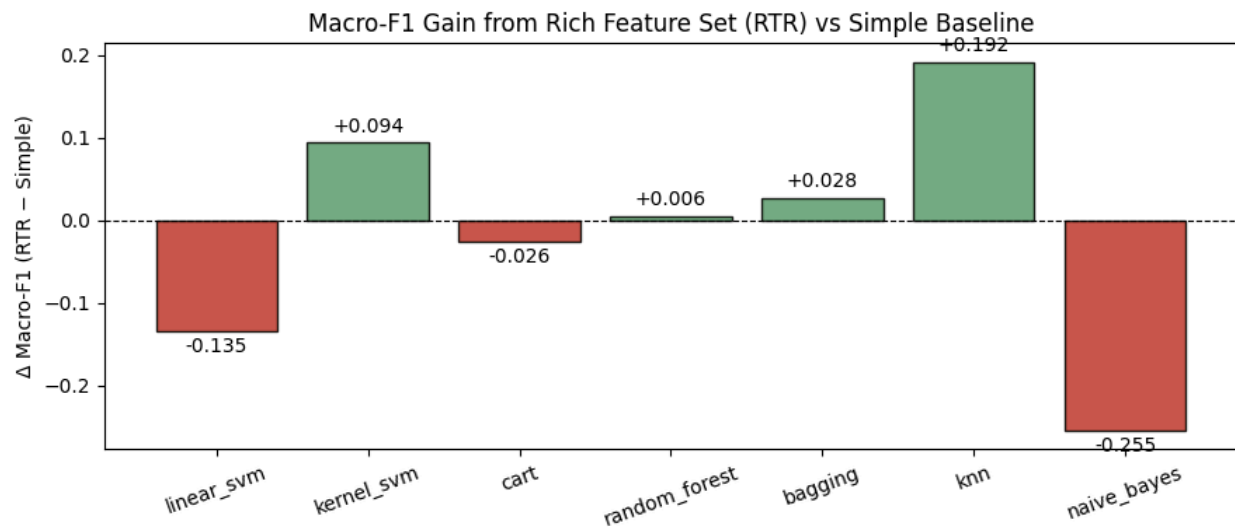


Figure 3: Macro-F1 Gain from Rich Feature Set (RTR) vs Simple Baseline

Phase II Outputs — Saved for Phase III

All Phase II outputs were persisted for use in Phase III:

- Fit probability scores (per observation, per class) for ensemble construction in Phase III
- Saved model objects for comparison against simple benchmark models

Phase III — Size Recommender System

Objective

Translate fit probabilities from Phase II into actionable size recommendations for new customer-product pairs. The recommender answers the final business question: given what we know about a customer's body profile and what similar customers have experienced, what size should this person order?

Architecture

Collaborative Filtering Framework

A sparse user-item matrix is constructed where rows represent customers and columns represent category-size combinations (e.g., dress_8, top_4). Cell values are fit probabilities derived from Phase II ensemble predictions. Cosine similarity measures fit behavior similarity between customers, enabling the system to identify relevant neighbors for any query.

Similarity-Weighted Prediction

For each recommendation request, the system identifies the top-k most similar users and aggregates their fit probabilities for each candidate size within the requested category. More similar users contribute more heavily to the prediction via similarity weighting, producing a score representing the likelihood of each size fitting the target customer.

Cold-Start Handling

For new customers without purchase history, the system performs cold-start matching by identifying the training user with the most similar body measurements (height, weight, age, bust band, cup size) and using that user's interaction profile as a proxy. A category-specific popularity prior is blended into predictions to stabilize recommendations when neighbor sets are small.

Performance Results

Evaluation Setting	Hit@1	Hit@3	Interpretation
Within RTR (user-level split)	3.0%	9.1%	Cold-start dominant; sparsity limits collaborative signal
Cross-retailer (ModCloth → RTR)	10.5%	32.2%	Category-level transfer generalizes well across brands

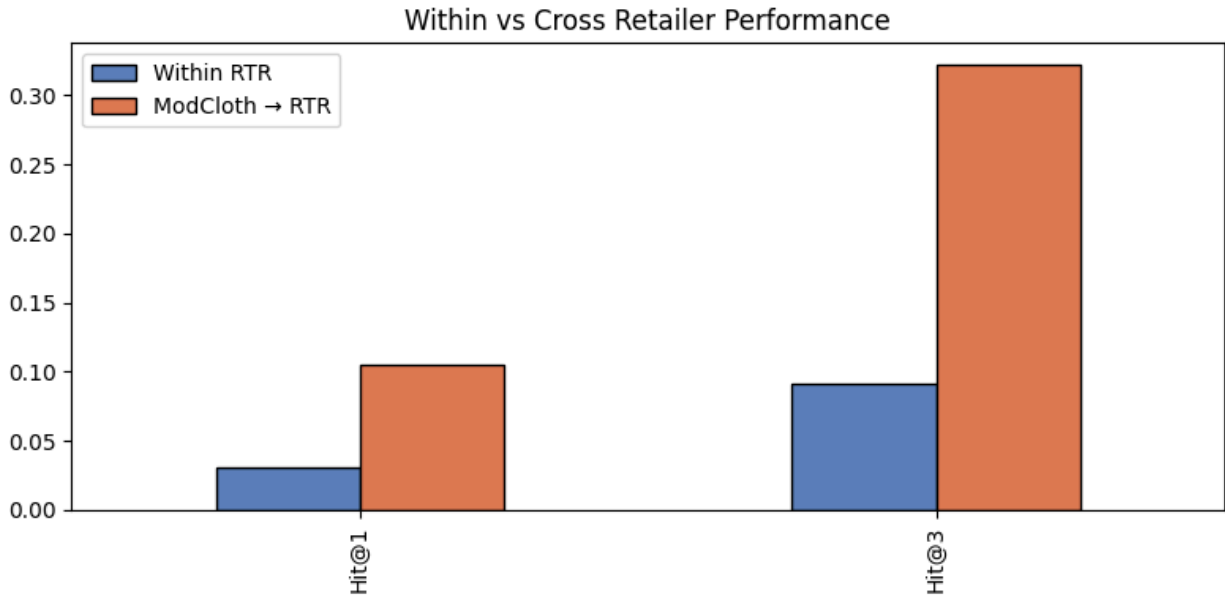


Figure 4: Within vs Cross Retailer Performance

The cross-retailer result is particularly meaningful from a business perspective: size patterns learned from one retailer's customers generalize to another when evaluated at the category level. This demonstrates that the underlying body-type and fit relationships captured in Phase I and Phase II are not retailer-specific artifacts — they reflect genuine regularities in how body measurements relate to garment fit.

Limitations & Future Work

Data Limitations

- Reviews are self-reported; fit labels reflect subjective perception, not objective measurement
- Size inconsistency varies across manufacturers, complicating cross-brand comparison
- Missing clothing dimension data limits measurement-gap feature engineering

Modeling Limitations

- No temporal dimension: the pipeline does not account for how fit perception may shift across seasons or over a customer's purchase history
- KNN, Kernel SVM, Linear SVM, and Naive Bayes were tuned on subsamples due to computational constraints while full-data tuning may yield further performance gains

Recommender Limitations

- Within-retailer performance is constrained by sparsity; most customers have a single transaction, which limits collaborative filtering signal
- Wide variety of sizes recommended due to lack of size consistency across the catalog
- RTR subcategories were grouped into broader categories to align with ModCloth, which may reduce garment-specific fit information; this loss of granular data has impacted the precision of our recommendations

Scope Limitations

- Analysis is limited to women's clothing on RentTheRunway and ModCloth which means findings may not generalize to men's sizing or different retail categories
- Pipeline does not account for body build nuances (e.g., athletic vs. pear-shaped within the same measurement profile)

Recommended Next Steps

- Temporal modeling: incorporate purchase history sequences to capture how fit preference evolves
- Richer garment features: integrate product dimension data (e.g., brand-published measurements) to enable measurement-gap features at scale
- Expand to men's and plus-size: the pipeline architecture is category-agnostic and can be retrained on new populations
- Deploy LDA feedback loop: use real-time review text to update topic features and retrain classifiers periodically

Conclusion

This project demonstrates that clothing fit prediction is a tractable machine learning problem only when the full dimensionality of fit failure is captured. Fit fails because body measurements are continuous and overlapping, because garment context shapes fit independently of body shape, because customers at the edges of standard sizing are systematically underserved, and because subjective fit perception leaves signals in language that structured data cannot encode. The three-phase pipeline addresses each of these failure modes in sequence: Phase I discovers the latent structure, Phase II learns to predict from it, and Phase III converts predictions into recommendations.

The most important finding is that the richness of the feature engineering, driven by Phase I unsupervised analysis, was responsible for the largest performance gains. KNN's accuracy improvement from 72% to 91% is not a property of KNN; it is a property of the features. This means the primary investment for any retailer looking to deploy this system is not in modeling complexity but in data richness: collecting body measurements, review text, and garment context at scale. The pipeline is designed to reward that investment.

Bottom Line for Stakeholders

A customer who receives the right size on the first order is a customer who stays. The pipeline built here, from clustering to understand who customers are, classification to predict what will fit them, and collaborative filtering to recommend it, provides the technical infrastructure to make that happen at scale.

Citation

Misra, R., Wan, M., & McAuley, J. (2018). *Decomposing Fit Semantics for Product Size Recommendation in Metric Spaces*. In Proceedings of the 12th ACM Conference on Recommender Systems, pp. 422–426.

Misra, R., & Grover, J. (2021). *Sculpting Data for ML: The First Act of Machine Learning*. ISBN 9798585463570.